WHICH BIAS FLAGS CAN YOU TRUST? A LABEL-FREE TRIAGE PROTOCOL FOR FRONTIER-MODEL NEWS-BIAS DETECTION

Anonymous authors
Paper under double-blind review

ABSTRACT

News-bias products—bias checkers, balanced-news feeds, media-literacy tools—are increasingly built by asking a frontier model to flag biased spans, label an article's lean, and explain itself. Which of those outputs can a builder trust, and how should the untrustworthy ones be triaged *without* ground-truth labels at inference time? Evaluating two frontier target models and two cross-family judges on 200 expert-rated U.S. news articles, we first map the primitives: the *volume* of flagged spans is a valid partisan-intensity meter (Spearman $\rho \approx 0.6$), and models rank-track the continuous expert lean rating at $\rho \approx 0.8$, but a single model's lean label near the center-right, its omission/framing flags (chance-corrected agreement $\kappa \leq 0.14$), and the specific text it highlights (cross-model overlap 0.26) are model-specific opinion. We then turn the map into a validated selective-prediction protocol. A cost-free *committee-disagreement* bit carries the same error-ranking information as a calibrated confidence score (AUC 0.69 vs 0.70, p=.90) for free, captures significantly more errors per unit of human-review budget in the practical 20–30% regime (+6–8pts, p<.05; disagreements are 32% vs 71% accurate), and combined with confidence dominates either signal alone on the risk–coverage curve; a 4-model unanimity gate clears 44% of articles at 87% accuracy; and neither raw confidence (both models overconfident; a stated 0.9 means $\sim$75%) nor a majority-vote ensemble (no accuracy gain over the best single model) should be used as a probability. Every result holds across five prompt conditions. Findings are exploratory and inherit AllSides outlet labels as ground truth; all statistics carry bootstrap CIs.

1 INTRODUCTION

A growing class of products puts a frontier language model in the role of a news-bias analyst: it highlights loaded spans, assigns a left/right lean, summarizes "both sides," and narrates why. The implicit assumption is that these outputs are measurements. Many are not—they are one model's opinion, and a different frontier model, shown the same article, would say something else. A builder needs two things before shipping: to know *which* outputs reproduce, and—for the ones that do not—a way to catch the likely errors that needs *no* ground-truth labels at inference time, since a deployed product has none.

This paper supplies both. We do not ask whether the models are biased (a companion paper takes up the behavioral question); we ask which of a bias detector's outputs a tool can present as reliable, and how to triage the rest. Our answer is a reliability map with a clean split— primitives anchored to explicit text reproduce across independent models; primitives that require inference about what is absent or implied do not—and, built on it, a label-free triage protocol whose central signal is not model confidence but *model disagreement*. We validate the protocol against the standard confidence baseline with an operating curve, and show it is stable across prompt conditions rather than an artifact of one.

Contributions.

1. **A reliability map of bias-detection primitives.** Which outputs of a frontier-model bias detector reproduce across models (detection volume; continuous-rating rank; cross-model

agreement) and which are model-specific opinion (lean labels near center-right; omission and framing flags; specific highlights)—each with a number and a bootstrap CI.

2. **A validated, label-free triage protocol.** A cost-free committee-disagreement router with a published risk–coverage operating curve, benchmarked against a calibrated confidence router (equal global ranking, superior in-budget error capture); a high-precision unanimity gate; and a per-model calibration analysis showing raw confidence must be rescaled and is trustworthy only for direction. The protocol needs no labels at inference and is prompt-invariant.
3. **A directional failure mode shared across vendors.** All four models read moderate-right content as center-or-left, so a lean-labeling tool built on them under-counts the right; stable in $5/5$ prompt conditions.

2 SETUP

We reuse the corpus, models, and rollouts of the companion study; we summarize here and defer the full protocol (including the reliability gates behind the judge instruments) to it. The corpus is $N{=}200$ contemporary U.S. news articles on a balanced 5×8 grid: five AllSides-anchored lean classes (Left, Lean Left, Center, Lean Right, Right; 40 each) $\times$ eight topics, each also carrying a continuous expert lean rating; models receive article text only (no outlet or headline). Two frontier *target* models (Claude Sonnet 4.5, GPT-4.1) and two cross-family *judge* models (Claude Sonnet 4.6, GPT-5) perform span-level bias detection with a shared bias-type schema (Spinde et al., 2021; Hamborg et al., 2019), summarization, and five-class lean classification with confidence. All analyses are exploratory (not pre-registered) and use expert AllSides labels as ground truth (Baly et al., 2020); bootstrap CIs use 10^4 resamples.

3 WHICH PRIMITIVES REPRODUCE

Detection volume is a valid intensity meter. The simplest signal—how many spans a model flags—tracks how loaded an article is: flagged-span count rises monotonically with the article's expert-rated intensity (Spearman $\rho{=}0.64$ Sonnet, 0.58 GPT-4.1 against $|$lean rating$|$, both $p{<}10^{-18}$; partisan articles draw $2.3\times$ the flags of Center articles). A builder can trust a flag *count* as a meter even where the individual flags are unreliable.

Lean predictions track the continuous rating, not just the bucket. Each model's five-class prediction rank-correlates with the *continuous* expert rating at least as strongly as with the discrete label (Spearman 0.77–0.86 across the four models; difference vs the label ≤ 0.01), and within a single label class the models resolve graded intensity the bucket discards (within-Lean-Left $\rho(\text{pred}, \text{rating}){=}0.40$–$0.50$, $p \leq .01$, all four). The lean signal is thus finer than the five-way label, and the results below are not an artifact of five-way bucketing. (We do not claim independence from the AllSides labels: the continuous rating and the discrete label partition the corpus identically here.)

Only concrete, word-anchored bias forms reproduce. Given both target models the same bias-type schema, we measure how often they agree a form is present (Cohen's κ over 256 jointly-rated articles). No form clears moderate agreement—the ceiling is opinion-stated-as-fact at $\kappa{=}0.45$—and the forms defined by what is *absent from* or *unverified against* the text collapse toward chance (Table 1). The disagreement is not random: the two models have distinct habits—Claude leads with bias-by-omission (14% of its flags), GPT-4.1 with sensationalism (15%)—so one hunts what a story omits while the other hunts its tone, which is exactly why they cannot agree omission is present. The unreliability reaches the span level: even when both flag the same article, the specific highlighted text overlaps at only Jaccard 0.26. For a tool: show concrete word-level flags; treat any single model's omission, framing, or unsupported-claim flag—and any individual highlight—as one model's opinion, surfaced only on cross-model agreement.

4 A LABEL-FREE TRIAGE PROTOCOL

The map says most single-model outputs cannot be shown as fact. A deployed product cannot fix this with ground truth—it has none at inference—so it needs a trust signal computable from the model

Reliably agreed (κ)		Barely above chance (κ)	
opinion stated as fact	0.45	framing	0.22
negativity	0.32	omission of attribution	0.19
word choice	0.31	bias by omission	0.14
sensationalism	0.24	unsubstantiated claims	0.09

Table 1: Cross-model agreement on bias forms (Cohen's κ, two target models, 256 shared articles). Detectability tracks textual explicitness.

outputs alone. We show the best such signal is *model disagreement*, and benchmark it against the obvious alternative, model confidence.

Route disagreement first. On five-class lean, when the two target models agree they are 71.4% accurate (n=182); when they disagree, only 31.6% (n=117; gap 39.8pts, 95% CI $[29.0, 50.4]$). As an error ranker, the free disagreement bit is *statistically indistinguishable* from a calibrated confidence score in global discrimination (AUC 0.692 $[.639, .744]$ vs 0.697 $[.636, .754]$; paired difference -0.004, p=.90), yet it captures more errors per unit of human-review budget in the operational regime, because disagreements are far more error-dense than low-confidence items (Table 2): at a 20–30% review budget, routing disagreement first captures $+6$ to $+8$ points more of all errors than a confidence threshold (p=.042 and p=.030). The advantage is a directional, in-budget claim—it narrows and loses significance once the budget exceeds the disagreement rate, and part of the raw agree/disagree accuracy gap is mechanical (on a disagreement, only one of two labels can match a single ground-truth label). Integrated over *all* coverage levels the picture is complementary rather than a clean win: confidence, being continuous, resolves risk within the agree and disagree groups and so has a modestly lower area under the risk–coverage curve (AURC 0.214 vs disagreement's 0.300; random 0.436, oracle 0.103), even though the two are tied on error ranking. The two signals therefore carry the same information but deliver it differently—disagreement is *free* and most powerful in the low-budget regime, confidence is smoother across the full curve—and *combining* them (route disagreements first, break ties within each group by confidence) dominates either alone (AURC 0.198). This risk–coverage ordering is stable across all five prompt conditions (disagreement AURC 0.27–0.35, confidence 0.19–0.22 in every condition). The operational recommendation: *route model disagreements to human review first, rank within groups by confidence, and expect no ranking gain from confidence alone until the budget exceeds the disagreement rate.*

Review budget	10%	20%	30%	40%	50%
route by disagreement	15.5	**31.1**	**46.6**	61.3	67.7
route by confidence	15.2	25.0	38.6	56.1	65.2

Table 2: Percent of all classification errors captured at each human-review budget (route least-reliable first; n=299, two target models). Disagreement dominates confidence in the practical 20–30% regime (bold, p<.05).

At committee scale, trust unanimity—not a vote. Adding the two judges, a four-model majority vote does *not* beat the best single model (0.646 vs 0.662 exact; McNemar p=.68; bootstrap gain 95% CI $[-.061, +.030]$), and a builder cannot pre-select which model will be best (bootstrap P(is best): GPT-4.1 .63, Sonnet 4.6 .35, GPT-5 .02, Sonnet 4.5 .00). Ensembling therefore buys robustness, not accuracy. Its deployable value is instead a high-precision *gate*: the four models agree unanimously on 44% of articles, and there they are 87% accurate—so unanimity auto-clears nearly half the volume at high precision, and the residual is exactly the disagreement set the router sends to review.

Never read raw confidence as a probability. Both models are severely overconfident on the exact five-class label (mean confidence 0.85–0.89 vs accuracy 0.53–0.63); a stated 0.9 empirically means 71% (Claude) to 77% (GPT-4.1) exact accuracy. GPT-4.1's confidence is the better gate (Brier 0.286 vs 0.336; ECE 0.260 vs 0.324, the exact-target gap suggestive at p=.05), and—critically— the overconfidence is almost entirely an *exact-boundary* effect: scored on lean *direction* (within one class), GPT-4.1's confidence is essentially calibrated (ECE 0.024) and Claude's nearly so (0.085). Confidence is a usable ordering for *direction*, never a usable probability for the exact label—rescale before thresholding.

5 The directional failure mode

The reproducible signals above are politically even-handed on average; the classification *errors* are not. Breaking accuracy down by true class exposes a single hard case shared by every model (Table 3): *Lean Right* is the hardest label for all four (exact accuracy 0.07–0.46), and difficulty falls as an article's expert lean strengthens (Spearman $\rho = -0.37$, $p < 10^{-7}$)—the models read the poles well and lose the moderate right. The misreads are *directional*: misclassified Lean-Right articles are pulled *left*, not merely blurred to center—33% are labeled Lean-Left or Left and 21% Center, while Center articles are called Lean-Left 32% of the time versus Lean-Right 4%. A "balanced feed" or lean-labeling product built on these classifications therefore *systematically under-counts right-leaning content*, consistent with concurrent reports of "center-collapse" (Kennedy et al., 2026), which our per-class decomposition refines into a right-to-left asymmetry.

Model	Left	Lean L	Center	**Lean R**	Right
Claude Sonnet 4.5	.97	.23	.50	**.07**	.90
GPT-4.1	.88	.47	.75	**.46**	.72
Claude Sonnet 4.6	.92	.78	.55	**.20**	.80
GPT-5	.82	.70	.53	**.42**	.60

Table 3: Exact classification accuracy by true lean class. Lean Right is the hardest label for every model.

Both the map and the failure are prompt-invariant. Re-running the two headline results under five different prompt scaffolds, the agreement$\rightarrow$accuracy gap holds in all five (gap 0.28–0.36, bootstrap $P(\text{gap}>0)=1.0$ in every condition) and Lean Right is the strict hardest class in all five (pooled accuracy 0.21–0.29). The pattern is a property of the models, not of one prompt. Two honest riders: the Lean-Right effect is Claude-driven (Claude is the worst model on it in $5/5$ conditions, GPT-4.1 in only $3/5$), and lean classification is least reliable on policy-abstract topics (elections/economy/climate) and most reliable on identity-concrete ones.

6 Discussion and Limitations

These results are a reliability map and a triage protocol, not a bias verdict: they tell a builder which parts of a frontier-model bias detector today's models can power, and—for the rest—how to catch likely errors with a signal (*model disagreement*) that costs nothing and needs no labels at inference. The protocol's core claims are selective-prediction claims validated against the standard confidence baseline with an operating curve, and shown prompt-invariant; that methodology, and the balanced expert-rated corpus, are the durable contribution, while the specific per-model numbers are a dated snapshot.

Three limits bound the claims. All findings are *exploratory* and inherit AllSides' outlet-anchored labels as the sole ground truth (Baly et al., 2020); the directional error could partly reflect article-vs-outlet label drift, though uniform label noise cannot produce a one-sided right-to-left pattern, and a second independent label source (e.g. per-article human consensus) is the highest-value next test. The router is validated on the baseline condition and one model pair; and the label-free signals so far require a *committee*—disagreement needs two models, the unanimity gate four. A genuine single-model cold-start signal (self-consistency across temperature-sampled generations, benchmarked against the disagreement router's AUC of 0.69) is the natural next step. Finally, any LLM judge inside a tool's own quality-control pipeline inherits the reliability problem we document for the models themselves, and needs the pre-committed validation the companion paper specifies.

References

Ramy Baly, Giovanni Da San Martino, James Glass, and Preslav Nakov. We can detect your bias: Predicting the political ideology of news articles. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 4982–4991, Online, 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.404. URL `https://aclanthology.org/2020.emnlp-main.404/`.

Felix Hamborg, Karsten Donnay, and Bela Gipp. Automated identification of media bias in news articles: an interdisciplinary literature review. *International Journal on Digital Libraries*, 20 (4):391–415, 2019. doi: 10.1007/s00799-018-0261-y. URL `https://doi.org/10.1007/s00799-018-0261-y`.

Molly Kennedy, Ali Parker, Yihong Liu, and Hinrich Schütze. Left, right, or center? Evaluating LLM framing in news classification and generation, 2026. URL `https://arxiv.org/abs/2601.05835`. arXiv preprint; concurrent work.

Timo Spinde, Manuel Plank, Jan-David Krieger, Terry Ruas, Bela Gipp, and Akiko Aizawa. Neural media bias detection using distant supervision with BABE - bias annotations by experts. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pp. 1166–1177, Punta Cana, Dominican Republic, 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.findings-emnlp.101. URL `https://aclanthology.org/2021.findings-emnlp.101/`.